Example 1.13

Find adjoint of $A = \begin{bmatrix} 1 & -2 & -3 \\ 0 & 1 & 0 \\ -4 & 1 & 0 \end{bmatrix}$

Solution

$A_{ij} = (-1)^{i+j} M_{ij}$

$A_{11} = (-1)^{1+1} M_{11} = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} = 0$

$A_{12} = (-1)^{1+2} M_{12} = -\begin{vmatrix} 0 & 0 \\ -4 & 0 \end{vmatrix} = 0$

$A_{13} = (-1)^{1+3} M_{13} = \begin{vmatrix} 0 & 1 \\ -4 & 1 \end{vmatrix} = 0-(-4) = 4$

$A_{21} = (-1)^{2+1} M_{21} = -\begin{vmatrix} -2 & -3 \\ 1 & 0 \end{vmatrix}$

$= -(0-(-3)) = -3$

$A_{22} = (-1)^{2+2} M_{22} = \begin{vmatrix} 1 & -3 \\ -4 & 0 \end{vmatrix}$

$= 0-12 = -12$

$A_{23} = (-1)^{2+3} M_{23} = -\begin{vmatrix} 1 & -2 \\ -4 & 1 \end{vmatrix}$

$= -(1-8) = 7$

$A_{31} = (-1)^{3+1} M_{31} = \begin{vmatrix} -2 & -3 \\ 1 & 0 \end{vmatrix}$

$= 0-(-3) = 3$

$A_{32} = (-1)^{3+2} M_{32} = -\begin{vmatrix} 1 & -3 \\ 0 & 0 \end{vmatrix}$

$= 0-0 = 0$

$A_{33} = (-1)^{3+3} M_{33} = \begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix}$

$= 1-0 = 1$

$[A_{ij}] = \begin{bmatrix} 0 & 0 & 4 \\ -3 & -12 & 7 \\ 3 & 0 & 1 \end{bmatrix}$

Adj $A = [A_{ij}]^T$

$= \begin{bmatrix} 0 & -3 & 3 \\ 0 & -12 & 0 \\ 4 & 7 & 1 \end{bmatrix}$

1.2.4 Inverse of a matrix

Let A be any non-singular matrix of order n. If there exists a square matrix B of order n such that $AB = BA = I$ then, B is called the inverse of A and is denoted by A^{-1}

We have A (adj A) = (adj A)A

$= |A|\, I$

$\Longrightarrow A\left(\frac{1}{|A|} adj\ A\right) = \left(\frac{1}{|A|} adj\ A\right)A$

$= I$

$\Longrightarrow AB = BA = I$ where $B = \frac{1}{|A|} adjA$

$$A^{-1} = \frac{1}{|A|} adjA$$

NOTE

(i) If B is the inverse of A then, A is the inverse of B
i.e., $B = A^{-1} \Rightarrow A = B^{-1}$

(ii) $AA^{-1} = A^{-1}A = I$

(iii) The inverse of a matrix, if it exists, is unique.

(iv) Order of inverse of A will be the same as that of A.

(v) $I^{-1} = I$, where I is the identity matrix.

(vi) $(AB)^{-1} = B^{-1}A^{-1}$ (vii) $|A^{-1}| = \frac{1}{|A|}$

DO YOU KNOW?

Let A be a non-singular matrix, then

$$A^2 = I \Leftrightarrow A = A^{-1}$$

Example 1.14

If $A = \begin{bmatrix} 2 & 4 \\ -3 & 2 \end{bmatrix}$ then, find A^{-1}.

Solution

$A = \begin{bmatrix} 2 & 4 \\ -3 & 2 \end{bmatrix}$

$|A| = \begin{vmatrix} 2 & 4 \\ -3 & 2 \end{vmatrix} = 16 \neq 0$

01_11th_BM-STAT_Ch-1-EM.indd 9 21-04-2020 12:19:54 PM